

Web Development

NCSC101

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What is Web Development?

- Process of building and maintaining websites and web applications.
- Covers frontend (client-side), backend (server-side), and databases.
- Enables communication between users and applications.



Evolution of the Web

- Web 1.0: Static pages (1990s)
- Web 2.0: Interactive and user-generated content (2000s)
- Web 3.0: Decentralized, AI-driven (present)



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Key Insight

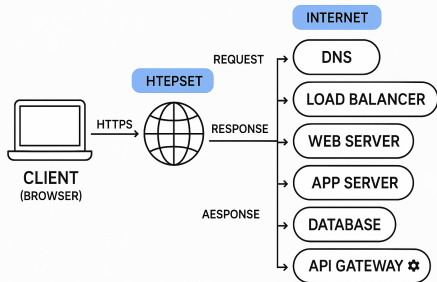
Client-server model is foundational to all.



Client-Server Architecture

Definition

A model where the client (user's device) requests resources or services from a server (remote computer).





Client-Server Model

- A design pattern for distributed applications.
- Client: Requests data/services (browser, mobile app).
- Server: Provides data/services (web server, database).
- Communication via Internet protocols (HTTP/HTTPS).



How Client-Server Communication Works

1. User enters a URL in the browser (client).
2. Browser sends a request to the server.
3. Server processes the request.
4. Server sends response (HTML, JSON, etc.).
5. Browser displays the content to the user.



Web Protocols

- **HTTP:** HyperText Transfer Protocol, stateless communication.
- **HTTPS:** Secure version of HTTP using SSL/TLS encryption.
- **REST APIs:** Communication using standard HTTP methods (GET, POST, PUT, DELETE).



Frontend vs Backend

Frontend (Client-Side):

- Technologies: HTML, CSS, JavaScript
- Focus: User interface, interactivity, responsiveness

Backend (Server-Side):

- Technologies: Node.js, Python (Flask/Django), PHP, Java, .NET
- Focus: Data processing, business logic, databases



Introduction to Static Web Pages

What are Static Web Pages?

Web pages that are delivered to the user exactly as stored, without dynamic content generation. Built primarily with HTML and CSS.

Advantages

- Fast loading
- Easy to host
- No server-side processing needed



Introduction to Static Web Pages

Limitations

- No user-specific content
- Requires manual updates



What is HTML?

HyperText Markup Language

The standard markup language for creating web pages. Describes the structure of a web page using tags.

Key Fact

HTML is not a programming language; it's a markup language.

Versions

Current: HTML5, which adds semantic elements, multimedia support, etc.



HTML Document Structure

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Document Title</title>
</head>
<body>
  <!-- Content here -->
</body>
</html>
```

Explanation

- **<!DOCTYPE>**: Declares document type
- **<html>**: Root element
- **<head>**: Metadata
- **<body>**: Visible content



Basic HTML Tags

Common Tags

- `<h1>` to `<h6>`: Headings
- `<p>`: Paragraphs
- `<div>`: Divisions/sections
- `<span>`: Inline containers
- `<br>`: Line breaks
- `<hr>`: Horizontal rules

Example

```
<div><h1>Main Title</h1><p>Some text.</p></div>
```



Headings and Paragraphs

Headings

Use `<h1>` for main title, `<h2>` for subsections, down to `<h6>`.

```
<h1>Heading 1</h1>
```

```
<h2>Heading 2</h2>
```

Paragraphs

`<p>` tags for blocks of text.

```
<p>This is a paragraph with some text.</p>
```



Lists in HTML

Unordered Lists

`<ul>` with `<li>` items.

```
<ul>
  <li>Item 1</li>
  <li>Item 2</li>
</ul>
```

Ordered Lists

`<ol>` with `<li>` items.

```
<ol>
  <li>First</li>
  <li>Second</li>
</ol>
```




Links and Anchors

Hyperlinks

Use `<a>` tag with href attribute.

```
<a href="https://example.com">Visit Example</a>
```

Anchors

For internal links: `<a href="#section">Jump</a>` and
`<div id="section">Target</div>`



Images in HTML

Embedding Images

`<img>` tag with `src`, `alt`, `width`, `height` attributes.

```

```

Best Practice

Always include alt text for accessibility.



Tables in HTML

Structure

`<table>`, `<tr>`for rows, `<th>`for headers, `<td>`for data.

```
<table>
  <tr>
    <th>Header1</th>
    <th>Header2</th>
  </tr>
  <tr>
    <td>Data1</td>
    <td>Data2</td>
  </tr>
</table>
```



Forms Basics in HTML

Creating Forms

`<form>` tag with inputs. (Static pages can't process, but structure is HTML.)

```
<form action="/submit" method="post">  
  <label for="name">Name:</label>  
  <input type="text" id="name" name="name">  
  <input type="submit" value="Submit">  
</form>
```



What is CSS?

Cascading Style Sheets

Language for styling HTML documents: colors, layouts, fonts, etc.

Separation of Concerns

HTML for structure, CSS for presentation.

Versions

Current: CSS3, with modules for advanced features.



CSS Syntax

Basic Rule

Selector property: value;

```
p {  
  color: blue;  
  font-size: 16px;  
}
```

Comments

```
/* This is a comment */
```



Ways to Apply CSS

Inline

Directly in HTML: `<p style="color: red;">Text</p>`

Internal

In `<style>` tag in `<head>`.

External

Link to .css file: `<link rel="stylesheet" href="styles.css">`

Preference

Use external for maintainability.



CSS Selectors

Types

- Element: `p { }`
- Class: `.classname { }`
- ID: `#idname { }`
- Attribute: `[type="text"] { }`
- Pseudo-classes: `a:hover { }`



Element Selector

The element selector targets all HTML elements of a specific type.

Example

```
p {  
  color: blue;  
  font-size: 16px;  
}
```

This selects all `p` elements and applies blue color and 16px font size. **Sample HTML:**

This paragraph will be blue.



Class Selector

The class selector targets elements with a specific class attribute.

Example

```
.classname {  
background-color: yellow;  
padding: 10px;  
}
```

This selects all elements with `class="classname"` and applies a yellow background with padding. **Sample HTML:**

This div has a yellow background.



ID Selector

The ID selector targets a unique element with a specific ID attribute.

Example

```
#idname {  
font-weight: bold;  
color: red;  
}
```

This selects the element with `id="idname"` and makes the text bold and red.

Sample HTML:

This heading is bold and red.



Attribute Selector

The attribute selector targets elements with a specific attribute and value.

Example

```
input[type="text"] {  
border: 2px solid green;  
width: 200px;  
}
```

This selects all elements with **type="text"** and applies a green border and fixed width. **Sample HTML:**



Pseudo-classes Selector

Pseudo-classes target elements in a specific state, like hover or focus.

Example

```
a:hover {  
  color: purple;  
  text-decoration: underline;  
}
```

This changes the color to purple and adds an underline when hovering over elements. **Sample HTML:**

Hover over this link.



Colors and Backgrounds

Colors

Named (red), Hex (`#FF0000`), RGB (`rgb(255,0,0)`), HSL.

```
body {  
  background-color: #f0f0f0;  
  color: #333;  
}
```

Backgrounds

Images: `background-image: url('bg.jpg');`



Fonts and Text Styling

Fonts

```
font-family: 'Arial', sans-serif;
```

Text Properties

```
font-size, font-weight, text-align, text-decoration, line-height.
```

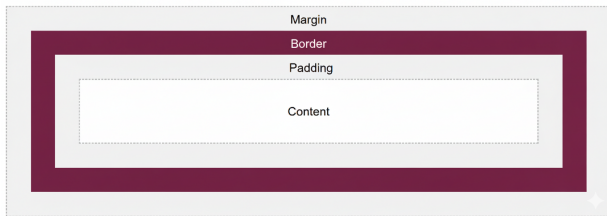
```
h1 {  
  font-family: 'Georgia', serif;  
  text-align: center;  
}
```



CSS Box Model

Components

Content, Padding, Border, Margin.



```
div {  
  margin: 10px;  
  padding: 20px;  
  border: 1px solid black;  
}
```

Figure: Box Model Diagram



Layout Techniques

Float

For simple layouts: float: left/right;

Flexbox

Modern: display: flex; on container.

```
.container {  
  display: flex;  
  justify-content: space-between;  
}
```

Grid

For complex: display: grid;



Best Practices and Tools

Best Practices

- Semantic HTML
- Mobile-first design
- Validate code (W3C validators)
- Optimize images

Tools

- Editors: VS Code, Sublime
- Browsers Dev Tools
- Frameworks: Bootstrap for quick styling



Web Development

Thank You for Listening!